A Focused Perspective on 3D Generative Models

Amine Maazizi
ENSTA Paris
amine.maazizi@ensta.fr

Abstract

Recent progress in image- and text-conditioned 3D generation is often presented as a sequence of increasingly capable models. A more useful interpretation is that the field has repeatedly changed *where its prior knowledge comes from*. Earlier reconstruction systems relied primarily on explicit geometric structure and native 3D supervision. Neural implicit representations and differentiable rendering then made it possible to optimize 3D scenes from image-space losses. Large pretrained image–text and diffusion models pushed this shift further by supplying semantic and appearance priors learned from web-scale 2D data, enabling 3D generation with little or no paired text–3D supervision. This focused perspective organizes representative work around four transitions: from explicit to learned geometry, from native 3D supervision to differentiable image supervision, from image supervision to frozen generative priors, and from unconstrained 2D guidance to mechanisms that restore 3D consistency. The resulting view is that modern 3D generation is not a replacement of geometry by foundation models. Rather, semantics increasingly come from large 2D priors, while geometry, multi-view constraints, retrieval, and reconstruction architectures determine whether those priors can be converted into coherent and usable 3D assets.

1 Introduction

Generating or reconstructing a 3D object from images or language is fundamentally underconstrained. A single image hides geometry, text specifies semantics without spatial measurements, and native 3D datasets remain far smaller than the image and image–text corpora used to train modern foundation models. The central question is therefore not only *which 3D representation should be used*, but also *which source of prior knowledge can constrain the missing geometry*.

The literature suggests a recurring redistribution of that prior. Early deep reconstruction methods encoded geometry directly through mesh connectivity or learned continuous shape fields [1–3]. Neural rendering then made 2D observations sufficient to optimize continuous 3D scene representations, with NeRF providing the clearest demonstration that a scene could be learned through differentiable rendering alone [4]. The arrival of CLIP and large diffusion models changed the source of supervision again: instead of asking only whether a rendered view matched an observed image, optimization could ask whether it matched the semantic and visual distribution learned by a large pretrained 2D model [5–7].

This shift enabled text-to-3D and stronger single-image reconstruction, but it also exposed a structural mismatch. A 2D generative model can judge or synthesize plausible views without representing a single persistent 3D object. Consequently, much of the subsequent literature can be read as an effort to recover the consistency that was weakened when supervision moved from 3D into 2D: staged optimization, redesigned score distillation, camera-conditioned multi-view generation, explicit 3D priors, retrieval, and feed-forward reconstruction all constrain the same ambiguity from different directions [8–13].

2 Geometry

2.1 Representations

Before large generative priors entered the pipeline, reconstruction quality depended heavily on how geometric structure was represented. Mesh-based methods such as Pixel2Mesh++ make this explicit: connectivity is fixed or inherited from a template, while learned graph operations predict deformations from image evidence [1]. This gives the model a strong surface prior, but also restricts topology and ties prediction to the chosen parameterization.

Implicit neural representations relaxed this constraint. DeepSDF represents shape through a continuous signed-distance function, while Occupancy Networks predict whether spatial locations lie inside or outside a surface [2, 3]. The important transition is not simply from meshes to multilayer perceptrons. It is from storing geometry explicitly to representing geometry as a queryable function whose parameters can be learned and optimized. This makes resolution less directly coupled to a discrete grid and creates a representation naturally compatible with gradient-based reconstruction.

These methods still rely heavily on 3D supervision or shape collections. Their contribution to later generative systems is therefore mainly representational: they provide flexible spaces in which geometry can be optimized, but they do not yet solve the scarcity of semantic 3D supervision.

2.2 Rendering

A second transition occurs when the renderer itself becomes part of the learning system. If a 3D state $\mathcal{S}$ can be rendered from a camera Θ_v through

$$I_v = \mathcal{R}(\mathcal{S}; \Theta_v), \quad (1)$$

then losses defined on images can update the underlying 3D parameters. NeRF operationalized this idea with a continuous radiance field optimized from posed multi-view images [4]. The key conceptual change is that explicit mesh or voxel labels are no longer required: multi-view photometric agreement becomes the supervisory signal.

This rendering bridge becomes crucial later because it allows any sufficiently differentiable image-space prior to supervise a 3D representation. Once gradients can pass from rendered pixels to scene parameters, the source of supervision no longer needs to be another 3D object.

3 2D Priors

3.1 Semantic Priors

CLIP-style image–text representations provided an early route from natural language to 3D optimization. Text2Mesh, for example, modifies the appearance and local geometry of an existing mesh so that rendered views align with a text prompt in a shared vision–language embedding space [5, 6]. The geometric object remains explicit, but the semantic objective is imported from a model trained outside the 3D domain.

Diffusion models make this transfer substantially stronger. Their denoising networks encode a rich image prior rather than only an embedding similarity. DreamFusion showed that such a frozen 2D diffusion model could supervise a 3D representation through Score Distillation Sampling (SDS), allowing text-conditioned 3D optimization without paired text–3D training data [7]. Schematically, the update takes the form

$$\nabla_\theta \mathcal{L}_{\mathrm{SDS}} \approx \mathbb{E}_{t,\epsilon}\left[w(t)\big(\hat{\epsilon}_\phi(x_t, y, t) - \epsilon\big)\frac{\partial x}{\partial \theta} \right], \tag{2}$$

where the frozen diffusion model supplies a denoising direction and the renderer transfers that direction to the 3D parameters θ.

The significance of DreamFusion is therefore broader than its particular NeRF pipeline. It demonstrates that large-scale 2D generative knowledge can substitute for paired 3D semantic supervision. The 3D representation becomes the object being optimized, while much of the semantic and appearance knowledge resides in the frozen image model.

3.2 2D–3D Gap

This transfer is powerful but incomplete. A diffusion model is trained to model images, not persistent objects across arbitrary viewpoints. Consequently, independently plausible views can correspond to incompatible geometry. Janus-like duplications, implausible back sides, oversmoothed geometry, and long per-instance optimization are manifestations of the same mismatch: the semantic prior is strong, but its native domain is not 3D.

Magic3D addresses part of this difficulty through staged representations: coarse neural-field optimization establishes global structure before a higher-resolution mesh stage refines detail [8]. ProlificDreamer instead revisits the distillation objective itself, replacing deterministic SDS with variational score distillation to improve fidelity and diversity [9]. These methods differ technically, yet both preserve the same overall paradigm: a frozen 2D generative prior remains the semantic engine, while changes to optimization and representation make that supervision better behaved in 3D.

4 Consistency

4.1 Multi-View

One response to the 2D–3D mismatch is to modify the image-generation stage so that viewpoint is no longer implicit. Zero-1-to-3 conditions a diffusion model on relative camera transformations, using large-scale image priors while explicitly learning novel-view transformations from synthetic data [10]. This reframes single-image reconstruction: instead of inferring the full object directly, a model first predicts consistent-enough observations from additional viewpoints, which can then supervise a downstream 3D reconstruction process.

Later systems push this idea toward direct and efficient image-to-3D pipelines. Unique3D generates structured multi-view images and normals before reconstructing a mesh, shifting much of the burden from slow per-instance SDS optimization to a learned multi-view generation and reconstruction pipeline [13]. Hi3D uses a related observation from another domain: temporal consistency learned by video diffusion can be repurposed as geometric consistency when the "video" follows an orbital camera trajectory around an object [14]. In both cases, 3D consistency is encouraged before final reconstruction by constraining the family of views supplied to it.

4.2 Guidance

A second response is to preserve the expressive 2D model while adding external geometric guidance. Sculpt3D retrieves semantically related 3D objects and uses sparse geometric supervision to constrain score-distilled generation without forcing dense imitation of the retrieved shape [11]. ReDream likewise treats retrieval as a way to improve score distillation: a relevant 3D asset supplies geometric information while the 2D diffusion model retains its broader appearance and semantic prior [12].

Retrieval is useful here because it occupies a middle ground between two extremes. Pure 2D distillation asks the optimizer to infer geometry from a prior that does not explicitly encode a persistent 3D instance. Fully learned native 3D generation instead requires enough 3D data to internalize geometric variation in model parameters. Retrieval provides non-parametric geometric memory on demand. Its weakness is equally clear: a semantically close reference can still have the wrong topology, pose, or part arrangement, so the system must control how strongly that reference constrains the output.

Phidias makes this control explicit by incorporating retrieved or user-provided 3D references into a reference-augmented diffusion model and dynamically modulating their influence [15]. The distinction from retrieval-augmented score distillation is important: in ReDream and Sculpt3D, retrieval primarily conditions an optimization process; in Phidias, the reference becomes part of the learned generative conditioning pathway. This suggests a broader transition from using external 3D information to stabilize optimization toward learning models that can consume 3D references directly.

5 Synthesis

5.1 Prior Source

The preceding methods are difficult to compare in a single benchmark-style taxonomy because they solve different tasks and occupy different positions in the pipeline. A more revealing distinction is the source and location of their prior knowledge. Geometry-first methods place strong assumptions in the representation or training data; neural rendering places supervision in observed views; SDS systems import semantic priors from frozen text-to-image diffusion; multi-view diffusion places additional consistency in the view generator; retrieval supplies instance-level geometric memory; and recent feed-forward systems increasingly internalize parts of the reconstruction pipeline into learned models.

Seen this way, the field is not moving monotonically from "3D methods" toward "2D methods." It repeatedly relocates information between the representation, the renderer, pretrained 2D models, retrieved 3D examples, and learned reconstruction networks. Improvements often come from choosing a better division of labor between these components.

5.2 Geometry

Large 2D priors are particularly effective at semantics, appearance, and text alignment because they inherit the scale and diversity of image-space training. Geometry contributes a different set of constraints: viewpoint persistence, spatial organization, surface extraction, editability, and compatibility with downstream graphics pipelines. The success of modern methods therefore comes less from replacing geometric inductive bias than from using it selectively to constrain stronger external priors.

Representation still affects what can be optimized and what can ultimately be delivered. Radiance fields provide a flexible differentiable optimization substrate but can be expensive to render and convert into editable assets. 3D Gaussian Splatting demonstrates that explicit anisotropic primitives can retain high rendering quality while enabling real-time novel-view synthesis [16]. Meshes remain attractive as deployable assets because they provide explicit surfaces, yet forcing optimization into mesh space too early can restrict topology or make gradients less forgiving. Hybrid pipelines consequently remain common: optimize or infer in a representation suited to learning, then reconstruct or refine a representation suited to use.

5.3 Feed-Forward 3D

SDS established that a frozen 2D model can supervise 3D without paired text–3D data, but it pays for that flexibility with optimization at inference time. Several later methods can be understood as attempts to amortize or remove this search. Camera-conditioned and multi-view diffusion models learn part of the missing 3D structure during training; feed-forward reconstruction networks convert generated views into geometry directly; and systems such as Meta 3D AssetGen target meshes, textures, and physically based materials rather than only visually convincing renderings [17].

This transition also changes what counts as success. A plausible rendered orbit is not equivalent to a usable asset. As pipelines become faster and more feed-forward, evaluation must increasingly account for geometry, topology, materials, relightability, and editability in addition to prompt fidelity or image similarity.

6 Open Questions

Three tensions follow directly from this perspective.

Learned vs. external 3D priors. Large native 3D datasets remain limited relative to image corpora, but multi-view synthetic data and increasingly large 3D repositories make stronger learned geometric priors possible. The open question is whether retrieval and hand-designed consistency mechanisms are temporary substitutes for future native 3D pretraining, or whether external geometric memory remains useful even when 3D models scale.

Joint evaluation. Image-space metrics and human preference can reward attractive views while missing non-manifold surfaces, duplicated geometry, poor topology, or unusable materials. Conversely, geometric distances may penalize plausible alternative shapes that satisfy a text prompt. Evaluation needs to reflect both perceptual plausibility and asset-level validity.

Consistency location. Current systems enforce consistency at different stages: in the 3D representation, across rendered views, inside a multi-view or video diffusion prior, through retrieved geometry, or during reconstruction. It remains unclear which constraints should be learned globally and which are better imposed explicitly at inference time. This design choice is likely to determine the balance among fidelity, controllability, speed, and generalization.

7 Conclusion

The progression from classical geometric reconstruction to modern generative 3D systems is best understood as a sequence of changes in supervision and inductive bias. Learned implicit fields made geometry differentiable and flexible; neural rendering moved supervision into image space; large vision–language and diffusion models supplied semantic priors learned far beyond available 3D datasets; and subsequent methods introduced multi-view generation, retrieval, explicit geometric guidance, and feed-forward reconstruction to recover the consistency that 2D priors do not provide by themselves.

The resulting systems are hybrids. Their semantic strength often originates in 2D pretraining, while their usefulness as 3D objects depends on how effectively geometric structure is represented, constrained, reconstructed, and exposed to downstream tools. Framing the literature around this division of labor provides a clearer connection between otherwise heterogeneous methods than a model-by-model comparison, and it highlights the central unresolved problem: how to retain the scale and expressiveness of 2D generative priors while making 3D consistency an intrinsic property rather than a correction applied afterward.

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